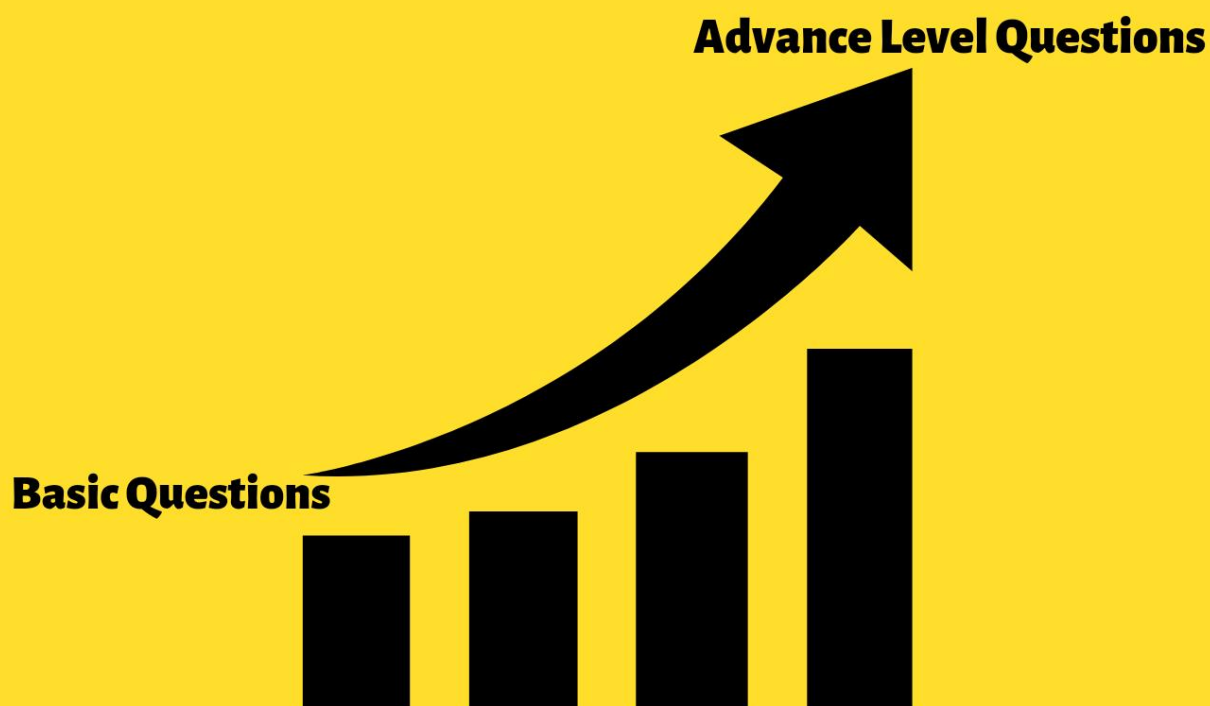


MATHS

COMPOUND INTEREST

ALL TYPE WISE QUESTIONS



Maths with Gajanand



COMPOUND INTEREST

Ex 1

Principal , Rate , Time , CI = ?
10,000 10% 2 years

m-I Basic:-

1st year :- 1000

2nd year :- $\frac{1000 \quad 100}{2000 + 100}$

CI = 2100 Ans

m-II

Principal $\swarrow$ $\searrow$ Amount
10,000 $\xrightarrow[=+1000]{+10\%}$ 11000 $\xrightarrow[+1100]{+10\%}$ 12100
C.I = 2100 Ans

m-III $10\% = \frac{1}{10}$

$\frac{10}{100} : \frac{11}{121}$
 $\swarrow \quad \searrow$
Principal $\ominus$ Amount
 $= 21$ C.I

C.I = $\frac{10,000}{100} \times 21$
 $= \underline{2100}$ Ans

m-IV

Formula

$$\text{Amount} = P \left(1 + \frac{r}{100} \right)^t$$

$$CI = P \left(1 + \frac{r}{100} \right)^t - P$$

$$C.I = 10,000 \left(1 + \frac{10}{100} \right)^2 - 10,000$$

$$CI = \boxed{2100} \text{ ans}$$

m-V

$$10\% = \frac{1}{10}$$

$$A = 10,000 \times \frac{11}{10} \times \frac{11}{10}$$

$$A = 12100$$

$$CI = A - P$$

$$CI = 12100 - 10,000$$

$$CI = \boxed{2100} \text{ ans}$$

m-VI

Effective rate Concept

$$\text{effective rate} = 21\%$$

$$C.I = 10,000 \times 21\%$$

$$C.I = \boxed{2100} \text{ ans}$$

Effective Rate

* For 2 years *

$$[CI = x + y + \frac{xy}{100}]$$

If Rate is same

$$= x + x + \frac{x^2}{100}$$

$$\boxed{\text{eff. Rate} = 2x + \frac{x^2}{100}}$$

$$\text{effective Rate} = 2x / x^2$$

$$2\% = 4/104 = 4.04\%$$

$$3\% = 6/109 = 6.09\%$$

$$4\% = 8/116 = 8.16\%$$

$$\star 5\% = 10/125 = \boxed{10.25\%}$$

$$6\% = 12/136 = 12.36\%$$

$$7\% = 14/149 = 14.49\%$$

$$8\% = 16/164 = 16.64\%$$

$$9\% = 18/181 = 18.81\%$$

$$\star 10\% = \frac{20}{210} = \boxed{21\%}$$

$$12\% = \frac{24}{254} = 25.44\%$$

$$15\% = \frac{30}{325} = 32.25\%$$

$$\star 20\% = \frac{40}{480} = \boxed{44\%}$$

$$30\% = \frac{60}{690} = 69\%$$

Ex.

<u>P</u>	<u>R</u>	<u>T</u>	C.I = ?
10,000	6%	2 year	

Solⁿ $10000 \times \frac{12.36}{100} = \boxed{1236}$ Ans

Ex.

<u>P</u>	<u>R</u>	<u>T</u>	C.I = ?
5000	15%	2 year	

Solⁿ $5000 \times \frac{32.25}{100} = \boxed{1612.5}$ Ans

Ex.

<u>P</u>	<u>R</u>	<u>T</u>	C.I
20,000	20%	2 years	?

Solⁿ $20,000 \times \frac{44}{100} = \boxed{8800}$ Ans

⇒ IF Rate is different

effective Rate = $x + y + \frac{xy}{100}$

Ex.

<u>P</u>	<u>R</u>	<u>T</u>	C.I
10,000	10%, 20%	2 years	?

Solⁿ $10,000 \times \frac{32}{100}$

eff. R = $10 + 20 + \frac{200}{100}$
= 32%

= $\boxed{3200}$ Ans

Ex. 2

Principal	Rate	Time	C.I
10,000	10%	3 years	??

Solⁿ:- Basic/Tree method:-

$$\text{I}^{\text{st}} \text{ yr} \rightarrow 1000$$

$$\text{II}^{\text{nd}} \text{ yr} \rightarrow 1000 \quad 100$$

$$\text{III}^{\text{rd}} \text{ yr} \rightarrow 1000 \quad 100 \quad 100 \quad 10$$

$$\text{C.I} = 3000 + 200 + 100 + 10$$

$$\text{C.I} = \boxed{3310} \text{ Ans}$$

m-II

$$10000 \xrightarrow{+10\%} 11000 \xrightarrow{+10\%} 12100 \xrightarrow{+10\%} 13310$$

Amount

$$\text{C.I} = \boxed{3310} \text{ Ans}$$

m-III $10\% = \frac{1}{10}$

$$\begin{array}{ccc} 10 & : & 11 \\ 10 & : & 11 \\ 10 & : & 11 \\ \hline \text{Principal} \rightarrow 1000 & : & 1331 \leftarrow \text{Amount} \\ & \swarrow & \searrow \\ & 331 & \leftarrow \text{C.I} \end{array}$$

$$\text{C.I} = \frac{10,000}{1000} \times 331$$

$$\text{C.I} = \boxed{3310} \text{ Ans}$$

Short

$$(10)^3 : (11)^3$$

$$1000 : 1331$$

$$\text{C.I} = \frac{10000}{1000} \times 331$$

$$= \boxed{3310}$$

m-V

$$C.I = P \left(1 + \frac{r}{100} \right)^3 - P$$

$$C.I = 10,000 \left(1 + \frac{10}{100} \right)^3 - 10,000$$

$$C.I = \boxed{3310} \text{ Ans}$$

m-v

$$10\% = \frac{1}{10}$$

$$\text{Amount} = 10,000 \times \frac{11}{10} \times \frac{11}{10} \times \frac{11}{10}$$

$$A = 13310$$

$$C.I = 13310 - 10,000$$

$$= \boxed{3310} \text{ Ans}$$

m-VI

Effective Rate:

$$10\% = \boxed{33.1\%}$$

$$C.I = 10,000 \times 33.1\%$$

$$= 10000 \times \frac{33.1}{100}$$

$$C.I = \boxed{3310} \text{ Ans}$$

Effective Rate

* For 3 years *

Rate :- $x\%$, $y\%$, $z\%$ for 3 years

$$\text{Eff. Rate} :- (x + y + z) + \frac{(xy + yz + zx)}{100} + \frac{xyz}{10000}$$

If Rate is same

$$\text{eff. R} :- 3x + \frac{3x^2}{100} + \frac{x^3}{10000}$$

$$3x / 3x^2 / x^3$$

$$2\% \quad :- \quad 6/12/08 \quad = \quad 6.1208\%$$

$$3\% \quad :- \quad 9/27/27 \quad = \quad 9.2727\%$$

$$* \boxed{5\%} \quad :- \quad 15/75/125 \quad = \quad \boxed{15.7625\%}$$

$$* \boxed{10\%} \quad :- \quad 30/300/1000 \quad = \quad \boxed{33.1\%}$$

$$15\% \quad :- \quad 52.0875\%$$

$$20\% \quad :- \quad 72.8\%$$

$$25\% \quad :- \quad 95.3125\%$$

$$30\% \quad :- \quad 114.7\%$$

Ex. $\begin{matrix} \boxed{P} \\ 5000 \end{matrix}$, $\begin{matrix} \boxed{R} \\ 10\% \end{matrix}$, $\begin{matrix} \boxed{\text{Time}} \\ 3 \text{ years} \end{matrix}$

Soln
 $5000 \times 33.1\%$
 $= \boxed{1655}$ Ans

Ex $\begin{matrix} \boxed{P} \\ 10000 \end{matrix}$ $\begin{matrix} \boxed{R} \\ 5\% \end{matrix}$ $\begin{matrix} \boxed{T} \\ 3 \text{ yr} \end{matrix}$

Soln
 $10000 \times \frac{15.7625}{100}$
 $= \boxed{1576.25}$ Ans

$\Rightarrow$ If Rates are different

Ex $P = 4000$
 $R = 2\% , 3\% , 5\%$
 $CI = ??$

Soln
 $\text{eff. Rate} = (x + y + z) + \frac{(xy + yz + zx)}{100} + \frac{xyz}{10000}$
 $= (10 + 0.31 + 0.0030)\%$
 $= 10.313\%$

$CI = 4000 \times 10.313\%$

$CI = \boxed{412.52}$ Ans

Ex. $\begin{matrix} \boxed{P} \\ 5000 \end{matrix}$ $\begin{matrix} \boxed{R} \\ 10\% \end{matrix}$, $\begin{matrix} \boxed{\text{Time}} \\ 3 \text{ years} \end{matrix}$

Soln
 $5000 \times 33.1\%$
 $= \boxed{1655}$ Ans

Ex $\begin{matrix} \boxed{P} \\ 10000 \end{matrix}$ $\begin{matrix} \boxed{R} \\ 5\% \end{matrix}$ $\begin{matrix} \boxed{T} \\ 3 \text{ yr} \end{matrix}$

Soln
 $10000 \times \frac{15.7625}{100}$
 $= \boxed{1576.25}$ Ans

$\Rightarrow$ If Rates are different

Ex $P = 4000$
 $R = 2\% , 3\% , 5\%$
 $CI = ??$

Soln
 $\text{eff. Rate} = (x + y + z) + \frac{(xy + yz + zx)}{100} + \frac{xyz}{10000}$
 $= (10 + 0.31 + 0.0030)\%$
 $= 10.313\%$

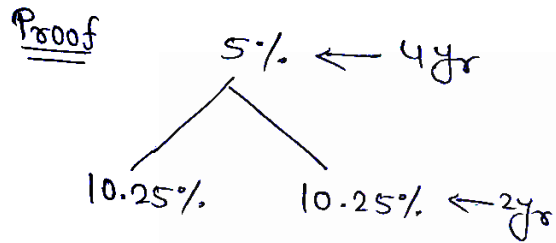
$CI = 4000 \times 10.313\%$

$CI = \boxed{412.52}$ Ans

Effective Rate

* For 4 years *

$$\begin{aligned} * 5\% &= 21.550625\% \\ * 10\% &= 46.41\% \end{aligned}$$



$$\Rightarrow A + B + \frac{AB}{100}$$

$$\Rightarrow 10.25 + 10.25 + \frac{10.25 \times 10.25}{100}$$

$$\Rightarrow 21.550625\%$$

<u>Ex</u>	<u>P</u>	<u>R</u>	<u>T</u>	<u>C.I</u>
	8000	10%	4yr	?

$$\Rightarrow 8000 \times 46.41\%$$

$$\Rightarrow \boxed{4512.8} \text{ Ans } \boxed{3712.8} \text{ Ans}$$

mf

$$A = 8000 \times \frac{11}{10} \times \frac{11}{10} \times \frac{11}{10} \times \frac{11}{10}$$

$$A = 11712.8$$

$$CI = 11712.8 - 8000 = \boxed{3712.8} \text{ Ans}$$

- 1] The compound interest earned in two years at 12% per annum Rs. 10176. What is the sum (in Rs) invested?

Soln:-

$$CI = 25.44\%$$

$$P = \frac{10176}{25.44} \times 100 = \boxed{40,000} \text{ Ans.}$$

m-II

$$12\% = \frac{3}{25}$$

$$(25)^2 : (28)^2$$

$$625 : 784$$

$$CI = 159$$

$$P = \frac{10176}{159} \times 625$$

$$= 64 \times 625$$

$$= \boxed{40,000} \text{ Ans}$$

- 2] A Certain sum will amount to ₹ 12,100 in 2 years at 10% per annum of compound interest, interest being compounded annually. The sum is:-

Soln:-

$$\text{eff. Rate} = 21\%$$

$$\text{Amount} = 121\% \longrightarrow 12100$$

$$1\% \longrightarrow 100$$

$$\text{Principal} = 100\% \longrightarrow \boxed{10,000} \text{ Ans}$$

m-II

$$(10)^2 : (11)^2$$

$$100 : 121 \longrightarrow 121,00$$

$$1 \longrightarrow 100$$

$$\boxed{10000} \text{ Ans}$$

3

What sum of money at compound interest will amount to Rs. 4,630.08 in three years if the rate of interest is 4% for the first year, 5% for the second year and 6% for the third year?

Soln (a) 4500 (b) 4800 (c) 4000 (d) 3500
[CHSL-2020]

$$P \times \frac{104}{100} \times \frac{105}{100} \times \frac{106}{100} = 4630.08$$

$$P \times \frac{104}{100} \times \frac{35}{100} \times \frac{106}{100} = 1543.36$$

digit sum:- $P \times 5 \times 8 \times 7 = 22$

$$P = \frac{4}{10} = 4 \quad \text{check option (c) Ans}$$

4 In how many years will a sum of Rs. 320 amount to Rs. 405 if interest is compounded at 12.5% per annum?

Soln:- Basic

$$320 \left(\frac{9}{8}\right)^t = 405$$

$$\left(\frac{9}{8}\right)^t = \frac{405}{320}$$

$$\left(\frac{9}{8}\right)^t = \frac{81}{64}$$

$$\left(\frac{9}{8}\right)^t = \left(\frac{9}{8}\right)^2$$

$$t = \boxed{2 \text{ years}} \text{ Ans}$$

m-II

$$320 : 405$$

$$\sqrt[2]{64} : \sqrt[2]{81}$$

$$8 : 9$$

$$\boxed{2 \text{ years}} \text{ Ans}$$

5] The Compound interest on Rs. 30,000 at 7% per annum for a certain time is Rs. 4,347. The time is -

Solⁿ:-

$$\frac{4347}{30,000} \times 100 = 14.49$$

↖ 7%
eff. Rate

2 years Ans

6] The amount on ₹25000 in 2 years at annually compound interest. If the rate for the successive years be 4% and 5% per annum respectively is -

Solⁿ:- $A = 25000 \times \frac{26}{25} \times \frac{21}{20} = \boxed{27300}$ Ans

m-II eff. Rate = $4 + 5 + \frac{20}{100} = 9.2\%$

$$A = 25000 \times \frac{109.2}{100} = \boxed{27300}$$
 Ans

7] The interest on ₹24000 in 2 years compounded annually when the rates are 8% p.a and 10% p.a for two successive years is ?

Solⁿ:- m-I eff. Rate

$$8 + 10 + \frac{80}{100} = 18.8\%$$

$$CI = 24000 \times \frac{18.8}{100} = \boxed{4512}$$
 Ans

$$A = 24000 \times \frac{27}{25} \times \frac{11}{10}$$

$$A = 28512$$

$$CI = 28512 - 24000$$

$$= \boxed{4512}$$
 Ans

8] The compound interest on Rs. 4000 after 3 year is Rs. 630.50. Then the rate of interest compounded yearly is:-

Soln:- eff. Rate = $\frac{630.50}{4000} \times 100$
 $= 15.7625$ Rate
5% Ans
for 3yr

9] A certain sum of money amounts to Rs. 2420 in 2 years and Rs. 2662 in 3 years at same rate of compound interest, compounded annually. The rate of interest per annum is -

Soln:-

$\begin{array}{c} \text{2y} \quad \quad \quad \text{1y} \\ \text{P} \quad \quad \quad 2420 \quad \quad \quad 2662 \end{array}$

$$\text{Rate \%} = \frac{2662 - 2420}{2420} \times 100$$

$$= \frac{242}{2420} \times 100 = 10\% \text{ Ans}$$

10] At what percent per annum will Rs. 3000 amounts to Rs. 3993 in 3 years if the interest is compounded annually?

Soln:-

$$3000 : 3993$$

$$\sqrt[3]{1000} : \sqrt[3]{1331}$$

$$10 : 11$$

$$\text{Rate} = \frac{1}{10} \times 100 = 10\% \text{ Ans}$$

III] The compound interest on a certain sum for two successive years are Rs. 225 and Rs 238.50 The rate of interest per annum is -

Soln:- Rate:- $\frac{238.5 - 225}{225} \times 100$
 $= \frac{13.5}{225} \times 100 = \boxed{6\%}$ Ans

[12] If the amount is $3\frac{3}{8}$ times the sum after 3 years at compound interest compounded annually, then the rate of interest per annum is -

Soln:- $3\frac{3}{8} \Rightarrow \frac{27}{8}$
 $\sqrt[3]{8} : \sqrt[3]{27}$
 $2 : 3$
 Rate% = $\frac{1}{2} \times 100 = \boxed{50\%}$ Ans

[13] On a certain Principal if the Simple Interest for two years is 4800 and Compound interest for two years is 5088, what is the rate of interest?

Soln:-
 I yr 2400 $\quad [5088 - 4800]$
 II yr 2400 288

Rate% = $\frac{288}{2400} \times 100$
 $= \boxed{12\%}$ Ans

14 What is the compound interest on a sum of ₹7200 for $2\frac{2}{5}$ years at 20% p.a., interest compounded yearly (nearest to an integer) ?

Soln:- **m-I**

effective Rate $20\% \times \frac{2}{5}$

20% 20% 8%

44% , 8%

$$\Rightarrow 44 + 8 + \frac{44 \times 8}{100}$$

$$\Rightarrow 55.52\%$$

$$CI = 7200 \times \frac{55.52}{100}$$

$$= \boxed{3997} \text{ Ans}$$

m-II

$$8\% = \frac{2}{25}$$

$$A = 7200 \times \frac{6}{5} \times \frac{6}{5} \times \frac{27}{25}$$

$$A = 11197.44$$

$$CI = 11,197.44 - 7200$$

$$= \boxed{3997.44} \text{ Ans}$$

15 The compound interest on a sum of Rs 20,000 at 15% p.a. for $2\frac{2}{3}$ years, interest compounded yearly is:

Soln:-

15% , 15% , 10% $\leftarrow 15 \times \frac{2}{3}$

P	A
20	23
20	23
10	11
4000	5819

Cr $\rightarrow$ 1819

$$CI = \frac{20,000}{4000} \times 1819$$

$$CI = \boxed{9095} \text{ Ans}$$

m-II

eff. Rate

15% 15% 10%

32.25% , 10%

$$\text{eff R} = 45.475\%$$

$$CI = 20,000 \times 45.475\%$$

$$= \boxed{9095} \text{ Ans}$$

[16] What is the compound interest on a sum of ₹10000 at 14% p.a. for $2\frac{5}{7}$ years where the interest is compound yearly? (nearest to ₹1)

Soln:-
 $14\% , 14\% , 10\%$
 $\quad \quad \quad \underbrace{\quad \quad \quad}$
 $\quad \quad \quad 29.96\% , 10\%$

eff Rate:- $29.96 + 10 + \frac{29.96 \times 10}{100}$

$= 39.96 + 2.996$

$= 42.956\%$

C.I. = $10,000 \times 42.956\%$

$= \boxed{4295.6}$ Ans

[17] The compound interest on a certain sum in $2\frac{1}{2}$ years at 10% p.a. interest compounded yearly is 1623. The sum is:

Soln:-
 $10\% , 10\% , 5\%$
 $\quad \quad \quad \underbrace{\quad \quad \quad}$
 $\quad \quad \quad 21\% , 5\%$

eff Rate :- $21 + 5 + \frac{105}{100}$

$= 27.05\%$

$P = \frac{1623}{27.05} \times 100$

$= \boxed{6000}$ Ans

m-II

10	11
10	11
<u>20</u>	<u>21</u>
P → 2000	2541 ← A
$\underbrace{\quad \quad \quad}$ C.I. → 541	

$P = \frac{1623}{541} \times 2000$

$= 3 \times 2000$

$= \boxed{6000}$ Ans

18 A certain sum amounts to ₹ 4,205.55 at 15% p.a. in $2\frac{2}{5}$ years, interest compounded yearly. The sum is?

Soln:-

15%, 15%, 6%

P	A
20	23
20	23
<u>50</u>	<u>53</u>
$20 \times 20 \times 50$	$23 \times 23 \times 53 \longrightarrow 4205.55$
	$\downarrow \text{Unit} \longrightarrow 0.15$

$$P = 20 \times 20 \times 50 \times 0.15$$

$$= \boxed{3000} \text{ Ans}$$

19 A sum amounts to Rs. 7727.104 at the rate of 12% per annum compounded annually after three years. What is the value of Principal?

Soln:- $12\% = \frac{3}{25}$

$$x \times \frac{28}{25} \times \frac{28}{25} \times \frac{28}{25} = 7727.104$$

$$x = \boxed{5500}$$

Ans

20 A invested an amount of x rupees in a Bank for 2 years which gave 5% interest in year 1 and 6% interest in year 2. The amount received after 2 years is Rs. 24,486. What is the value of x?

Soln:-

$$x \times \frac{21}{20} \times \frac{53}{50} = 24486$$

$$x = \boxed{22000} \text{ Ans}$$

m-II eff. Rate

$$5 + 6 + \frac{30}{100}$$

$$= 11.3\%$$

$$P = \frac{24486 \times 100}{111.3} = \boxed{22000} \text{ Ans}$$

Q1

Rs. 260200 is divided between Ram and Shyam so that the amount that Ram receives in 4 years is the same as that Shyam receives in 6 years. If the interest is compounded annually at the rate of 4% per annum then Ram's share is-

Soln

$$\text{Ram} \times \left(\frac{26}{25}\right)^4 = \text{Shyam} \times \left(\frac{26}{25}\right)^6$$

$$4\% = \frac{1}{25}$$

$$\frac{\text{Ram}}{\text{Shyam}} = \frac{676}{625}$$

$$\boxed{260200}$$

Ram : Shyam

676 : 625

$$\text{Ram} = \frac{260200}{1301} \times 676$$

$$= \boxed{135200} \text{ Ans}$$

[22] A sum of Rs. 12000 amounts to Rs. 20,736 in 3 years at a certain rate percent per annum interest compounded annually. What will amount of the same sum do in 2 years at the same rate on compound interest?

Soln:-

$$12000 : 20736$$

$$\sqrt[3]{12000} : \sqrt[3]{20736}$$

$$10 : 12$$

$$\text{Rate \%} = 20\%$$

$$\text{Amount} = 12000 \times \frac{6}{5} \times \frac{6}{5}$$

$$= \boxed{17280} \text{ Ans}$$

[23] A Sum of ₹15000 is lent at 16% p.a. compound interest. What is the difference between the compound interest for the second year and the third year?

Soln:-

$$CI \rightarrow 15000 \times \frac{4}{25} = 2400 \longrightarrow 2400 \times \frac{4}{25} = 24 \times 16$$

$$I \rightarrow 2400$$

$$II \rightarrow 2400 \quad 24 \times 16$$

$$III \rightarrow 2400 \quad 24 \times 16 \quad 24 \times 16 \quad 24 \times 16 \times \frac{4}{25}$$

$\underbrace{\hspace{10em}}_{CI_3 - CI_2}$

$$CI_3 - CI_2 = 24 \times 16 + 24 \times 16 \times \frac{4}{25}$$

$$= 384 + 61.44$$

$$= \boxed{445.44} \text{ Ans}$$

24) A sum of ₹ 18,000 is lent at 10% p.a. compound interest, compounded annually. What is the difference between the compound interest for 3rd year and 4th year?

Soln:-

$I \rightarrow 1800$

$$\pi \rightarrow 1800 \quad 180$$

III $\rightarrow$ 1800 180 180 18

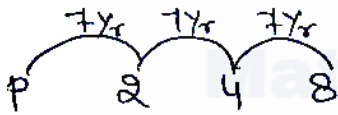
IV $\rightarrow$ 1800 180 180 180 18 18 18 1.8

$$C_{I_4} - C_{I_3} = 180 + 18 + 18 + 1.8$$

$$= \boxed{217.80} \text{ ms}$$

[25] If a certain sum becomes 2 times in 7 years at compound interest, then in how many years, it will become 8 times?

Soln

$$m-1$$


21 years

m-II

⑦ years $\longrightarrow$ 2 times

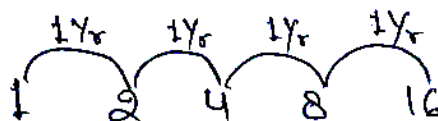
7 Years $\longrightarrow$ 8 times

26

$$3 \times 7 = \boxed{21 \text{ years}} \underline{\text{long}}$$

Q6 A sum of money becomes eight times in 3 years, if the rate is compounded annually. In how much time will the same amount at the same compound rate become sixteen times?

Soln:-



$$18 = 2^3$$

$14r \rightarrow 2 \text{ times}$

Time = 4 years

m-II

3 years $\longrightarrow$ $8 = 2^3$ times

① years $\longrightarrow$ 2 times

? Years $\longrightarrow$ 16 times

$\downarrow$
2④

$4 \times 1 = \boxed{4 \text{ years}}$ Ans

[27] A sum of Rs. 12,000, deposited at compound interest becomes double after 5 years. How much will it be after 20 years?

Soln:-

1 $\xrightarrow{5\text{yr}}$ 2 $\xrightarrow{5\text{yr}}$ 4 $\xrightarrow{5\text{yr}}$ 8 $\xrightarrow{5\text{yr}}$ 16

20 years $\longrightarrow$ 16 times

$12000 \times 16 = \boxed{1,92,000}$ Ans

[28] A sum of Rs. 2000 amounts to Rs. 4000 in two years at compound interest. In how many years does the same amount becomes Rs. 8000.

Soln:-

2000 $\xrightarrow[2\text{Yr}]{2 \text{ times}}$ 4000 $\xrightarrow[2\text{Yr}]{2 \text{ times}}$ 8000 = $\boxed{4 \text{ years}}$ Ans

m-II

2000 $\xrightarrow[2 \text{ times}]{\quad}$ 4000 $\longrightarrow$ ② years

2000 $\xrightarrow[4 \text{ times}]{\quad}$ 8000 $\longrightarrow 2 \times 2 = \boxed{4 \text{ years}}$ Ans
 $\downarrow$
2②

[29] An amount of money appreciates to Rs 7,000 after 4 years and to Rs. 10,000 after 8 years at a certain compound interest compounded annually. The initial amount of money was:-

Soln:-
$$P = \frac{7000 \times 7000}{10,000} = \boxed{\text{Rs. 4900}} \underline{\text{Ans.}}$$

[30] A sum of money invested at compound interest amounts to Rs. 650 at the end of first year and Rs. 676 at the second year. The sum of money is-

Soln:-
$$P = \frac{650 \times 650}{676} = \boxed{625} \underline{\text{Ans}}$$

[31] A sum amounts to ₹ 8,028 in 3 years and to ₹ 12,042 in ~ 6 years at a certain rate percent per annum, when the interest is compounded yearly. The sum is:-

Soln:-
$$P = \frac{8028 \times 8028}{12042} = \boxed{5352} \underline{\text{Ans}}$$

[32] A sum of money becomes Rs 11,880 after 4 years and Rs. 17,820 after 6 years on compound interest, if the interest is compounded annually. What is the half of the sum (in Rs)?

[SSC CGL mains 2020]

Soln

[29 Jan - 2022]

$$11880 \xrightarrow{2\text{Yr}} 17820$$

$$11880 : 17820$$

$$2 \text{ Years} \rightarrow 2 : 3$$

$$P \xrightarrow{2y} \xrightarrow{2y} 11880$$

$$P \times \frac{3}{2} \times \frac{3}{2} = 11880$$

$$P = 11880 \times \frac{2}{3} \times \frac{2}{3} = 5280$$

$$\begin{aligned} \text{half of Principal} &= \frac{5280}{2} \\ &= \boxed{2640} \text{ Ans} \end{aligned}$$

[33] The compound interest on 12,000 for 9 months at 20% Per annum, interest being compounded quarterly is?

Soln:-

P	R	T
12000	20%	9 month

$$\begin{aligned} &\downarrow \text{New} \\ &\frac{20\%}{12 \text{ month}} \times 3 \text{ month} \end{aligned}$$

$$\begin{array}{|c|c|c|} \hline 3m & 3m & 3m \\ \hline \end{array}$$

New Time

$$= \frac{20}{4} = \boxed{5\%}$$

$$\boxed{3 \text{ years}}$$

$$CI = 12000 \times \frac{15.7625}{100} \leftarrow \text{Effective Rate}$$

$$= \boxed{1891.5} \text{ Ans}$$

*

If Rate Given	New Time	New Rate
3 month	$T \times 4$	$R \times \frac{1}{4}$
6 month	$T \times 2$	$R \times \frac{1}{2}$
9 month	$T \times \frac{4}{3}$	$R \times \frac{3}{4}$
8 month	$T \times \frac{3}{2}$	$R \times \frac{2}{3}$

[34] What is the compound interest earned on Rs. 80,000 at 40% per annum in 1 year compounded quarterly?

Soln

P	New Time	Rate
80,000	1×4 = 4 years	$40 \times \frac{1}{4}$ = 10%

eff. Rate = 46.41%

$$CI = 80,000 \times \frac{46.41}{100} = \boxed{37128} \text{ Ans}$$

[35] The compound interest on ₹ 24000 at 10% per annum for $1\frac{1}{2}$ years, interest being compounded semi-annually is -

Soln

P	New Rate	New Time
24000	$10 \times \frac{1}{2}$ = 5%	$\frac{3}{2} \times 2$ 3 years

$$CI = 24000 \times 15.7625\%$$

$$= \boxed{3783} \text{ Ans}$$

[36] What will be the compound interest on a sum of ₹31,250 for 2 years at 12% p.a. if the interest is compounded 8-monthly.

Soln

P	New Rate	New Time
31250	$12 \times \frac{2}{3}$ = 8%	$2 \times \frac{3}{2}$ 3 years

$$\begin{array}{cc}
 P & A \\
 (25)^3 & (27)^3 \\
 15625 & 19683 \\
 \hline
 & CI = 4058
 \end{array}$$

$$8\% = \frac{2}{25}$$

$$CI = \frac{31250}{15625} \times 4058 = \boxed{8116} \text{ Ans}$$

[37] What is the compound interest of a sum of 8100 for $1\frac{1}{4}$ years at 8% per annum, if the interest is compounded 5-monthly?

Soln

[P]	[R]	[T]
8100	$\frac{8\%}{12m} \times 5m$	$\frac{(\frac{5}{4} \times 12) \text{ month}}{5 \text{ month}}$
	New = $\frac{10\%}{3}$	= 3 years

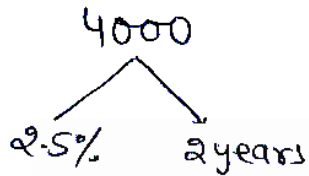
m-I Tree method

$$\begin{array}{l}
 I \rightarrow 270 \\
 II \rightarrow 270 \quad 9 \\
 III \rightarrow 270 \quad 9 \quad 9 \quad 0.3 \\
 \hline
 810 + 27 + 0.3 = \boxed{837.3} \text{ Ans}
 \end{array}$$

$$\left| \frac{10\%}{3} \rightarrow \frac{1}{30} \right|$$

[38] Rs. 4,000 is given at 5% per annum for one year and interest is compounded half yearly. Rs 2000 is given at 40% per annum compounded quarterly for 1 year. The total interest received is nearest to: [CHSL-2020]

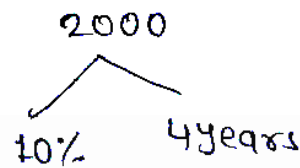
Soln:-



$$I \rightarrow 100$$

$$II \rightarrow \frac{100 \times 2.5}{200 + 2.5}$$

$$CI = \boxed{202.5}$$



$$CI = 2000 \times 46.41\%$$

$$= \boxed{928.2}$$

$$\text{Total CI} = 202.5 + 928.2 = \boxed{1130.7} \text{ Ans}$$

[39] A certain sum invested at 4% per annum compound interest, compounded half yearly, amounts to Rs. 7,803 at the end of one year. The sum is -

Soln:- **[m-I]**

$$\text{Rate} = \frac{4}{2} = 2\%$$

$$\text{Time} = 2 \text{ years}$$

$$CI = 4.04\%$$

$$\text{Sum} = \frac{7803}{104.04} \times 100$$

$$= \text{Rs. } \boxed{7500} \text{ Ans}$$

[m-II]

$$2\% \Rightarrow \frac{1}{50}$$

$$\begin{matrix} P & A \\ (50)^2 & (51)^2 \end{matrix}$$

$$2500 : 2601$$

↖ ↗
CI = 101

$$P = \frac{7803 \times 2500}{2601}$$

$$= \boxed{7500} \text{ Ans}$$

40 In what time will ₹ 64,000 amount to ₹ 68,921 at 5% per annum interest being compounded half yearly?

Soln:-

$$5\% \xrightarrow{\text{New}} \frac{5}{2}\%$$

$$\left| \frac{5}{2}\% \rightarrow \frac{1}{40} \right.$$

P

A

$$64000 : 68921$$

$$(40)^3 : (41)^3$$

$$\text{New time} = 3 \text{ year} \xrightarrow[\text{original Time}]{\div 2} \frac{3}{2} = \boxed{1\frac{1}{2} \text{ year}} \quad \underline{\underline{\text{Ans}}}$$

41 Find the rate percent per annum if ₹ 2,000 amounts to ₹ 2,315.25 in one and half years interest being compounded half yearly?

Soln:-

Basic

m-II

$$2315.25 = 2000 \left(1 + \frac{R}{2 \times 100} \right)^3$$

$$\frac{2315.25}{2000} = \left(1 + \frac{R}{200} \right)^3$$

$$\frac{9261}{8000} = \left(1 + \frac{R}{200} \right)^3$$

$$\frac{21}{20} = \left(1 + \frac{R}{200} \right)$$

$$R = \boxed{10\%} \quad \underline{\underline{\text{Ans}}}$$

$$\text{Rate} = \frac{315.25}{2000} \times 100$$

$$\text{New Rate} = 15.7625\%$$

↑
5% , for 3 years

original Rate

$$= 5 \times 2$$

$$= \boxed{10\%} \quad \underline{\underline{\text{Ans}}}$$

42 The effective annual rate of compound interest corresponding to a nominal rate of 15% per annum Payable half-yearly is-

Soln:- New rate , Time
 $\frac{15\%}{2}$ 2 years

Effective Rate:- $A + B + \frac{AB}{100}$

$$\Rightarrow 15 + \frac{225}{400} = \boxed{15.56\%} \text{ Ans}$$

43 A sum of Rs. 3,200 invested at 10% p.a. compounded quarterly amounts to Rs. 3,362. Compute the time period.

Soln:-

$$\begin{array}{l} 3200 : 3362 \\ 1600 : 1681 \end{array} \quad \left| \begin{array}{l} 10\% \xrightarrow{\text{New}} \frac{5\%}{2} = \frac{1}{40} \end{array} \right.$$

$$\Rightarrow (40)^2 : (41)^2$$

new time = 2 years

original Time = t

$$t \times 4 = 2$$

$$t = \boxed{\frac{1}{2} \text{ years}} \text{ Ans}$$

m-II $3362 = 3200 \left(1 + \frac{2.5}{100}\right)^{4t}$

$$\frac{1681}{1600} = \left(\frac{41}{40}\right)^{4t}$$

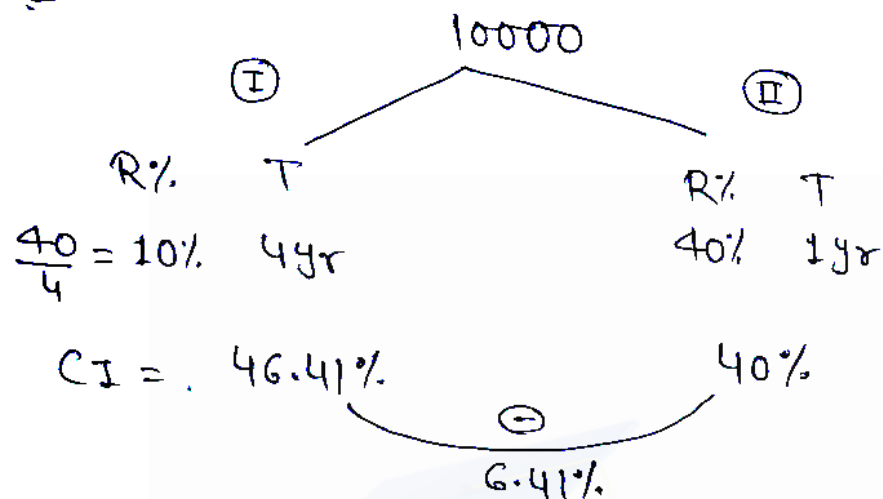
$$\left(\frac{41}{40}\right)^2 = \left(\frac{41}{40}\right)^{4t}$$

$$4t = 2$$

$$t = \boxed{\frac{1}{2} \text{ years}} \text{ Ans}$$

[44] What is the difference (in Rs) in Compound interest earned in 1 year on a sum of 10,000 at 40% per annum compounded quarterly and annually?

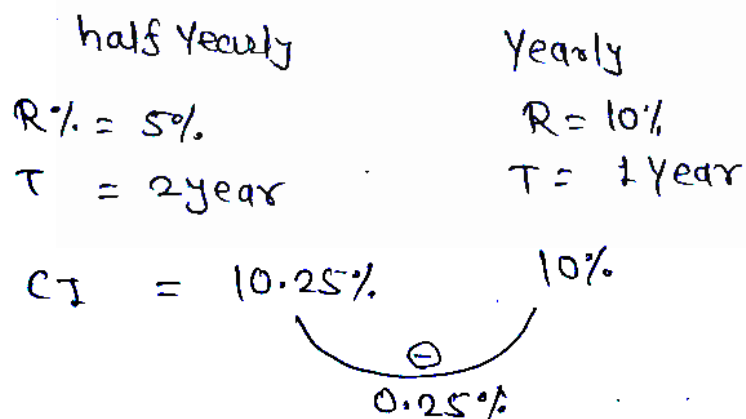
Soln:-



$$\text{Difference} = 10,000 \times 6.41\% = \boxed{641} \text{ Ans}$$

[45] The difference between the Compound Interest on a sum of Rs. 8,000 for 1 year at the rate of 10% per annum, interest compounded yearly and half yearly is:

Soln:-



$$\begin{aligned} \text{difference} &= 8000 \times 0.25\% \\ &= \boxed{\text{Rs. 20}} \text{ Ans} \end{aligned}$$

46 On what amount, compound interest compounded half-yearly is Rs. 1,545 at the rate of 12% per annum for 1 year?

Soln:- New Rate = $\frac{12}{2} = 6\%$

Effective rate = 12.36%

$$P = \frac{1545}{12.36} \times 100 = \boxed{12500} \text{ Ans}$$

47 If on a sum at 10% per annum, half yearly-compounded $1\frac{1}{2}$ year compound interest is Rs. 2,522, then the principal amount is:

Soln:- Rate = 5%

New Time = 3 year

Eff. Rate = 15.7625

$$P = \frac{2522}{15.7625} \times 100$$

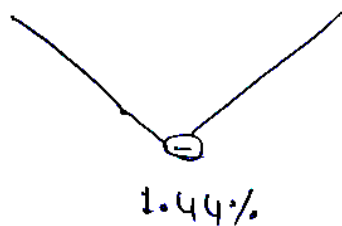
$$= \boxed{16000} \text{ Ans}$$

48 The difference between simple and compound interest compounded annually on a certain sum of money for 2 years at 12% per annum is Rs. 900. What is the value of given sum (in Rs.)?

Soln:-

$$CI = 25.44\%$$

$$SI = 24\%$$



$$P = \frac{900}{1.44} \times 100 = \boxed{62500} \text{ Ans}$$

[49] The difference between simple and compound interests compounded annually on a certain sum of money for 2 years at 9% per annum is Rs. 405. The sum is -

Soln:-

$$CI = 18.81\% \quad SI = 18\%$$

$$\underbrace{\hspace{1.5cm}}_{0.81\%}$$

$$P = \frac{405}{0.81} \times 100 = \boxed{50000} \text{ Ans}$$

[50] If the difference between the compound interest and simple at 17% on a sum money for 2 years (Compounded annually) is Rs 433.50, then the compounded interest (in Rs) is: [CPO-2020]

Soln:-

$$CI = 36.89\% \quad SI = 34\%$$

$$\underbrace{\hspace{1.5cm}}_{2.89\%}$$

$$CI = \frac{433.50}{2.89\%} \times 36.89$$

$$= 150 \times 36.89 \Rightarrow \boxed{5533.5} \text{ Ans}$$

[51] On a certain sum of money, the difference between the compound interest for a year, payable half-yearly, and the simple interest for a year is Rs. 180. If the rate of interest in both the cases is 10%, then the sum is -

Soln:-

5%, 2 Year

10%, 1 Year

$$CI = 10.25\%$$

$$SI = 10\%$$

$$CI - SI = 0.25\%$$

$$P = \frac{180}{0.25} \times 100 = \boxed{72,000} \text{ Ans}$$

[52] On a certain sum of money, the difference between the compound interest for a year, payable half-yearly, and the simple interest for a year is Rs. 56. If the rate of interest in both the cases is 16%, then the sum is -

Soln:- 8%, 2 year 16%, 1 year

$$CI = 16.64\% \quad SI = 16\%$$

$$CI - SI = 0.64\%$$

$$P = \frac{56}{0.64} \times 100 = \boxed{8750} \text{ Ans}$$

[53] What sum will give Rs. 244 as the difference between simple interest and compound interest at 10% in $1\frac{1}{2}$ years compounded half-yearly?

Soln:- 5%, 3 years 10%, $1\frac{1}{2}$ year

$$CI = 15.7625\% \quad SI = 15\%$$

$$CI - SI = 0.7625\%$$

$$P = \frac{244}{0.7625} \times 100 = \boxed{32,000} \text{ Ans}$$

[54] The difference of compound interest and simple interest for 3 years and for 2 years are in ratio 23:7 respectively, what is rate of interest per annum (in%)?

Soln

$$\begin{aligned} (CI - SI)_{\text{for 3 years}} &= P \left(\frac{r}{100} \right)^2 \left[\frac{300+r}{100} \right] & * \\ * & \\ * & (CI - SI)_{\text{for 2 years}} = P \left(\frac{r}{100} \right)^2 & * \end{aligned}$$

$$\frac{300+r}{100} = \frac{23}{7}$$

$$2100 + 7r = 2300$$

$$r = \boxed{\frac{200}{7}\%} \text{ Ans}$$

56 The difference between compound interest and Simple Interest on a sum for 2 years at 20% per annum is Rs 200. If the interest is compounded half yearly, then what is the difference (in Rs) between compound and simple interest for 1st year?

Soln:-

$$\left[P \left(\frac{R}{100} \right)^2 = CI - SI \right]$$

$$P \left(\frac{20}{100} \right)^2 = 200$$

$$P = 5000$$

New R = 10% , Time = 2 years

$$CI - SI = 5000 \left(\frac{10}{100} \right)^2 = \boxed{50} \text{ Ans}$$

m-II

CI	SI	CI-SI
44%	40%	4%

$$P = \frac{200}{4} \times 100 = 5000$$

New R = 10% Time = 2 years

CI	SI	CI-SI
21%	20%	1%

$$CI - SI = 5000 \times 1\% = \boxed{50} \text{ Ans}$$

Q6 The simple interest on a sum of money at 10% per annum for 2 years is Rs. 8,100. What would be the compound interest (in Rs.) on the same sum for the same period at 15% p.a. when the interest is compounded yearly?

Soln:- $SI = 20\%$

$$P = \frac{8100}{20\%} \times 100 = 40500$$

Now

$$P = 40500$$

$$R = 15\%$$

$$T = 2 \text{ year}$$

$$CI = ?$$

$$15\% = \frac{3}{20}$$

$$\begin{array}{cc} (20)^2 & (23)^2 \\ 400 & 529 \\ \hline & CI - 129 \end{array}$$

$$CI = \frac{40500}{400} \times 129 = \boxed{13061.25} \text{ Ans}$$

Q7 If the difference between the compound interest and simple interest on a certain sum of money for three years at 10% p.a. is Rs. 558. then the sum is?

soln:-

CI	SI	$CI - SI$
33.1%	30%	3.1%

$$P = \frac{558}{3.1} \times 100$$

$$= \boxed{18000} \text{ Ans}$$

Q-11

$$P \left(\frac{R}{100} \right)^2 \left(\frac{300+R}{100} \right) = CI - SI$$

$$P \times \frac{1}{100} \times \frac{310}{100} = 558$$

$$P = \boxed{18000} \text{ Ans}$$

[58] If the difference between CI and SI on a certain sum for three years is Rs. 8.181 at 3% per annum. Then find the principal.

Soln

$$CI = 9.2727\%$$

$$SI = 9$$

$$CI - SI = 0.2727\%$$

$$P = \frac{8.181}{0.2727} \times 100$$

$$P = \boxed{3000} \text{ Ans}$$

[59] Find the difference between the compound interest and the simple interest on 32,000 at 10% p.a. for 4 years.

Soln:-

$$CI = 46.41\%$$

$$SI = 40\%$$

$$CI - SI = 6.41\%$$

$$CI - SI = 32000 \times \frac{6.41}{100} = \boxed{\text{Rs. } 2051.20} \text{ Ans}$$

[60] The simple and compound interest that can be earned in two years at the same rate is Rs. 1,000 and Rs. 1,040 respectively. What is the rate (percent per annum) of interest?

Soln:- **m-I**

$$I \rightarrow 500$$

$$II \rightarrow 500 \quad 40$$

$$\text{Rate} = \frac{40}{500} \times 100$$

$$= \boxed{8\%} \text{ Ans}$$

m-II

$$CI - SI = 1040 - 1000$$

$$= 40$$

$$SI \text{ for 1 year} = 500$$

$$\text{Rate} = \frac{40}{500} \times 100$$

$$= \boxed{8\%} \text{ Ans}$$

[61] One certain Principal if the Simple interest for two years is Rs. 2400 and Compound interest for the two years is Rs. 2544, what is the rate of interest?

Soln:- $CI - SI = 2544 - 2400$
 $= 144$

SI for 2 years = 2400

$R\% = \frac{144}{2400} \times 100 = \boxed{12\%}$ Ans

[62] What is the rate of interest (in%) if simple interest earned on a certain sum for the 3 years is Rs. 6,000 and compound interest earned for 2 years is Rs. 4,160?

Soln:- m-I

I → 2000
 II → 2000 → 160
 III → 2000

$R\% = \frac{160}{2000} \times 100$
 $= \boxed{8\%}$ Ans

m-II

SI for 3 years = 6000

SI for 2 years = 4000

CI for 2 years = 4160

$CI - SI = 160$

$R\% = \frac{160}{2000} \times 100$
 $= \boxed{8\%}$ Ans

[63] A certain amount of money earns ₹540 as simple interest in 3 years. If it earns a compound interest of ₹376.20 at the same rate of interest in 2 years, find the Principal (in Rupees).

Soln I → 180
 II → 180 → 16.20
 III → 180

Rate = $\frac{16.20}{180} \times 100$
 $= 9\%$

$P = \frac{180}{9} \times 100 = \boxed{\text{Rs. 2000}}$ Ans

[64] On a certain sum of money, the simple interest for 2 years is Rs. 350 at the rate of 4% per annum. If it was invested at compound interest at the same rate for the same duration as before, how much more interest would be earned?

Soln:- SI eff Rate = $4\% \times 2 = 8\%$.

$$\text{Principal} = \frac{350}{8} \times 100 = 4375$$

$$\text{CI eff Rate} = 8.16\%$$

$$\text{CI-SI} = 0.16\%$$

$$= 4375 \times \frac{0.16}{100} = \boxed{7} \text{ Ans}$$

[65] If the simple interest of a certain sum at 7% per annum for 3 year is 9450. Then find the compound interest at 7% per annum for 2 years for that sum?

Soln:-

$$P = \frac{9450}{21\%} \times 100 = 45000$$

$$\text{CI} = 14.49\%$$

$$\text{CI} = \frac{45000 \times 14.49}{100}$$

$$\text{CI} = \boxed{6520.5} \text{ Ans}$$

[77-11]

$$45000 \times \frac{7\%}{100}$$

$$I \rightarrow 3150$$

$$\begin{array}{r} II \rightarrow 3150 \quad 220.5 \\ \hline 6300 + 220.5 \end{array}$$

$$= \boxed{6520.5} \text{ Ans}$$

[66] There is a 60% increase in an amount in 5 years at simple interest. What will be the compound interest on Rs. 6,250 for two years at the same rate of interest, when the interest is compounded yearly?

[CHSL-2020]

Soln:-

$$5 \text{ years} \longrightarrow 60\%$$

$$1 \text{ year} \longrightarrow 12\%$$

$$\text{Rate} = 12\%$$

$$P = 6250$$

$$T = 2 \text{ year}$$

$$CI = ?$$

[m-1] $CI = 25.44\%$

$$CI = 6250 \times \frac{25.44}{100}$$

$$CI = \boxed{1590} \text{ Ans}$$

$$\begin{matrix} P & A \\ (25)^2 & : (28)^2 \end{matrix}$$

$$\begin{matrix} 625 & : & 784 \\ \hline & 159 & \end{matrix}$$

$$12\% = \frac{3}{25}$$

$$CI = \frac{6250}{625} \times 159$$

$$= \boxed{1590} \text{ Ans}$$

[67] There is 40% increase in an amount in 8 years at simple interest. What will be the compound interest (in Rupee) of Rs. 30,000 after 2 years at the same rate?

Soln:- $R = \frac{40}{8} = 5\%$

$$CI = 10.25\%$$

$$CI = 30000 \times \frac{10.25}{100}$$

$$= \boxed{3075} \text{ Ans}$$

68 If in 3 years at simple interest the principal increases by 18%, what will be the compound interest (in Rs) earned on Rs. 25,000 in 3 years at the same rate?

Soln:- Rate = $\frac{18}{3} = 6\%$

P = 25000

T = 3 years

m-I
eff CI = $3x/3x^2/x^3$
= 18/108/216
= 19.1016%

CI = $25000 \times \frac{19.1016}{100}$
= Rs. 4775.4 Ans

m-II 25000 × 6% = 1500

1500 × 6% = 90

90 × 6% = 5.4

1500 × 3 + 90 × 3 + 5.4

4500 + 270 + 5.4

= 4775.4

69 In 4 years at simple interest the principal increases by 32%. What will be the compound interest earned (in Rs) on Rs. 24,000 in 3 years at the same rate?

Soln:- Rate = $\frac{32}{4} = 8\%$

P = 24000

T = 3 years

eff. Rate CI = $3x/3x^2/x^3$
= 24/192/512
= 25.9712%

CI = $24000 \times \frac{25.9712}{100} = \underline{623.3088} Ans$

70 A sum of Rs 10,500 amounts to Rs. 13,650 in 2 years at a certain rate percent annum simple interest. The same sum will amount to what in 1 year at the same rate, if the interest is compounded half yearly (nearest to Rs 1)

Soln

P	A	SI	Time
10,500	13,650	3150	2 year

$$\text{Rate} = \frac{3150}{10500 \times 2} \times 100 = 15\%$$

CI → New Rate, Time
 $\frac{15\%}{2}$, 2 year

$$A = 10500 \times \frac{43}{40} \times \frac{43}{40} = \boxed{12134} \text{ Ans}$$

$$\frac{15\%}{2} = \frac{3}{40}$$

71 A certain sum amount to Rs 15,500 in 2 years at 12% p.a. simple interest. The same sum will amount to what in $1\frac{1}{2}$ years at 10% p.a., if the interest is compounded half yearly (nearest to Rs 1)?

Soln

$$SI = 12\% \times 2 = 24\%$$

$$\text{Amount} = 124\%$$

$$P = \frac{15500}{124} \times 100 = 12500$$

CI:- Rate, Time
 5%, 3 years

$$(20)^3 : (21)^3$$

$$P \rightarrow 8000 : 9261 \leftarrow A$$

$$A = \frac{12500}{8000} \times 9261 = \boxed{14470} \text{ Ans}$$

72 The compound interest on a certain sum of money at 5% per annum for 2 years is Rs. 246. The simple interest on the same sum for 3 years at 6% per annum is -

Soln:-

$$P = \frac{246}{10.25\%} \times 100$$

$$P = 2400$$

$$SI = 2400 \times \frac{18}{100}$$

$$= \boxed{432} \text{ Ans}$$

m-II

$$CI = 10.25\%$$

$$SI = 18\%$$

$$SI = \frac{246}{10.25} \times 18$$

$$= \boxed{432} \text{ Ans}$$

73 A sum becomes Rs. 2916 in 2 years at 8% per annum compound interest. The simple interest at 9% per annum for 3 years on the same amount will be -

Soln:-

$$CI = 16.64\%$$

$$SI = 27\%$$

$$SI = \frac{2916}{116.64} \times 27 = \boxed{675} \text{ Ans}$$

74 At a certain rate per annum, the simple interest on a sum of money for one year is Rs. 260 and the compound interest on the same sum for two years is Rs. 540.80. The rate of interest per annum is -

Soln:-

$$\begin{array}{l} \text{m-I} \rightarrow 260 \\ \text{m-II} \rightarrow 260 \quad 20.8 \end{array}$$

$$\text{Rate} = \frac{20.8}{260} \times 100$$

$$= \boxed{8\%} \text{ Ans}$$

m-II

$$CI - SI = 540.80 - 260 = 20.8$$

$$\text{Rate} = \frac{20.8}{260} \times 100$$

$$= \boxed{8\%} \text{ Ans}$$

75 If the compound interest on a sum for 2 years at $12\frac{1}{2}\%$ per annum is Rs. 510, the simple interest on the same sum at the same rate for the same period of time is -

Soln:- $12\frac{1}{2}\% = \frac{1}{8}$

$$(8)^2 : (9)^2$$

$$64 : 81$$

$$\underbrace{\quad\quad\quad}_{17}$$

$$P = \frac{510}{17} \times 64$$

$$= 1920$$

$$SI = 1920 \times \frac{1}{8} \times 2 = \boxed{\text{Rs } 480} \text{ Ans}$$

76 The compound interest on a certain sum of money at a certain rate per annum for two years is Rs. 2050 and the simple interest on the same amount of money at the same rate for 3 years is Rs. 3000. Then the sum of money is -

Soln:-

$$SI \text{ for 3 years} = 3000$$

$$SI \text{ for 2 years} = 2000$$

$$CI \text{ for 2 years} = 2050$$

$$CI - SI = 2050 - 2000 = 50$$

$$\text{Rate \%} = \frac{50}{1000} \times 100$$

← SI of 1 year

$$= \boxed{5\%}$$

$$\text{Sum} = \frac{1000}{5} \times 100$$

$$= \boxed{20,000} \text{ Ans}$$

77 Simple interest on a certain sum of money for 3 years at 8% per annum is half the compound interest on Rs. 16000 for 2 years at 10% per annum. The sum placed on simple interest is -

Soln:- $SI = 8 \times 3$ $CI = 21\%$
 $= 24\%$

$$P \times 24\% = \frac{16000 \times 21\%}{2}$$

$$P = \boxed{\text{Rs } 7000} \text{ Ans}$$

78 Simple interest on a certain sum of money for 3 years at 18% per annum is half the compound interest on Rs. 9000 for 2 years at 10% per annum. The sum placed on simple interest is -

Soln:- $P \times 54\% = \frac{9000 \times 21\%}{2}$

$$P = \boxed{1750} \text{ Ans}$$

79 B borrows 5,000 from A at 6% p.a. simple interest and lends it to C at compound interest of 10% p.a. If B collects the money back from C after 2 years and repays A, the profit made by B in the transaction is -

Soln:-

SI	CI
12%	21%

$$\text{Profit} = CI - SI = 9\%$$

$$= 5000 \times \frac{9}{100} = \boxed{\text{Rs. } 450} \text{ Ans}$$

80 A man borrowed some money from a private Organisation at 5% simple interest per annum. He lended 50% of this money to another person at 10% Compound interest per annum and thereby the man made a Profit of Rs 3205 in 4 years. The man borrowed.

Soln:-

SI 4year

20%

CI 4year for 50%

$\frac{46.41\%}{2}$

$$\begin{aligned}\text{Profit} &= \frac{46.41\%}{2} - 20\% \\ &= \frac{6.41\%}{2}\end{aligned}$$

$$\text{Sum} = \frac{3205}{\frac{6.41\%}{2}} \times 100\% = \boxed{1,00,000} \text{ Ans}$$

81 A man borrowed some money from a private organization at 5% simple interest per annum. He lended this money to another person at 10% compound interest per annum, and made a profit of Rs. 26,410 in 4 years. The man borrowed,

Soln:-

SI

20%

CI

46.41%

$$\text{Profit } 26.41\% \longrightarrow 26410$$

$$1\% \longrightarrow 1000$$

$$\text{Sum} = 100\% \longrightarrow \boxed{1000000} \text{ Ans}$$

82 If the amount on a certain principal in 3 years at 12% rate of interest compounded annually is Rs. 12000 what will be the amount (in Rs.) after the 4th year?

Soln:-

$$12000 \times \frac{112}{100} = \boxed{\text{Rs. 13440}} \text{ Ans.}$$

83 If compound interest received on a certain amount in the 2nd year is Rs. 1200, what will be the compound interest (in Rs) for the 4th year on the same amount at 10% rate of interest?

Soln:-

1st year:- 1000

2nd year:- $1000 + 100 \Rightarrow 1100$

3rd year:- 1000 100 100 10

4th year:- $1000 + 100 + 100 + 100 + 10 + 10 + 10 + 1 \Rightarrow 1331$

$$\text{CI of 4th year} = \frac{1200}{1100} \times 1331 = \boxed{1452} \text{ Ans}$$

84 A, B and C invested amounts in the ratio 3:4:5 respectively. If the schemes offered compound at the rate of 20% per annum, 15% per annum and 10% per annum respectively, then what will be the ratio of their amounts after 1 year?

Soln:-

$$3 : 4 : 5$$

let

$$30 : 40 : 50$$

$$\begin{array}{ccc} +20\% \downarrow \downarrow & \downarrow \downarrow 15\% & \downarrow \downarrow 10\% \end{array}$$

Amount:- $36 : 46 : 55$ Ans

Q5 A borrowed ₹ 28,500 at 8% p.a. interest compounded annually. If ₹ 5,780 was paid at the end of first year, then the outstanding amount at the end of second year is:

Soln:-

$$\text{Principal} = 28500$$

$$\text{Interest} = 28500 \times \frac{8}{100} = 2280$$

$$\begin{aligned}\text{Amount} &= 28500 + 2280 \\ &= 30780\end{aligned}$$

$$\text{Remaining amount} = 30780 - 5780 = \boxed{25000} \text{ Ans}$$

Q6 A sum of ₹ 18,000 is invested for 16 months at 8% p.a. compounded half-yearly. What is the percentage gain at the end of 16 months, to the nearest whole number?

Soln:-

$$\text{Principal} = 18000$$

$$\begin{aligned}\text{New Rate} &= \frac{8}{12} \times 6 = 4\% \\ &= 4\%\end{aligned}$$

$$\text{New Time} = 32 \text{ months}$$

2 Years

8 month

$$\text{eff. Rate} = 8.16$$

$$\begin{aligned}&\frac{4\%}{12m} \times 8m \\ &= \frac{8\%}{3}\end{aligned}$$

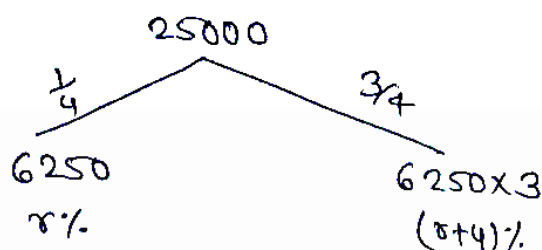
$$\text{eff. Rate} = 8.16 + \frac{8}{3} + \frac{8.16 \times 8}{3 \times 100}$$

$$= 8.16 + 2.66 + 0.21$$

$$\text{Gain \%} = \underline{\underline{11.03\%}} \text{ Ans}$$

- 87** A person invested one-fourth of the sum of ₹ 25,000 at a certain rate of simple interest and the rest at 4% p.a higher rate. If the total interest received for 2 years is ₹ 4,125, what is the rate at which the second sum was invested?

Soln:-



$$\frac{6250 \times r \times 2}{100} + \frac{6250 \times 3 \times (r+4) \times 2}{100} = 4125$$

$$\frac{6250}{100} \times 2 [r + 3(r+4)] = 4125$$

$$r = 5.25\%$$

$$\begin{aligned} \text{IInd Rate} &= (r+4)\% \\ &= (5.25 + 4) \\ &= \boxed{9.25\%} \end{aligned}$$

- 88** A certain amount grows at an annual interest rate of 12% compounded monthly. Which of following equation can be solved to find the number of years, y , that it would take for the investment to increase by a factor of 64?

(a) $64 = (1.01)^{12y}$ (b) $\frac{1}{64} = (1.04)^{12y}$ (c) $64 = (1.04)^{12y}$ (d) $8 = (1.01)^{6y}$

Soln:- Rate = 12% p.a = 1% per month

Time = $12y$ month

$$A = P \left(1 + \frac{R}{100}\right)^T$$

$$64 = 1 \left(1 + \frac{1}{100} \right)^{124}$$

$$64 = \left(\frac{101}{100} \right)^{124}$$

$$64 = (1.01)^{124} \quad \underline{\text{Ans}}$$

- [89]** A bank offers 5% compound interest per half year. A customer deposits Rs. 9600 each on 1st January and 1st July of a year. At the end of the year, The amount he would have gained by way of interest is:-

Soln

1 Jan $\rightarrow$ 30 June

1 July to 31 Dec.

$$P = 9600$$

rate = 5% half year

Time = 2 Year

$$\text{eff. Rate} = 10.25\%$$

$$\text{eff. Rate} = 5\%$$

$$\text{Total CI} = 9600 \times (10.25\% + 5\%)$$

$$= 9600 \times \frac{15.25}{100}$$

$$= \boxed{\text{Rs. 1464}} \quad \underline{\text{Ans}}$$

- [90]** A bank offers 20% compound interest per half year. A customer deposits Rs. 2800 each on 1st January and 1st July of a year. At the end of the year, the amount he would have gained by way of interest is —

- (a) Rs. 3584 (b) Rs. 896 (c) Rs. 1792 (d) Rs. 448

Soln:-

1 Jan - 30 Jun

1 July - 31 Dec

CI = 20% half yearly

↓

eff. Rate = 44%

eff. Rate = 20%

$$CI = 2800 \times \frac{(44+20)}{100}$$

$$= \boxed{1792} \text{ Ans}$$

[91] Rahul invested equal sums of money at compound interest under two schemes A and B. Under scheme A, the interest rate was 10% per annum and under scheme B, the interest rate was 12% p.a. The compound interest after two years on the sum invested in scheme A was Rs. 1050. How much is the interest earned under scheme B after two years, if the interest is compounded annually in both schemes? [SSC CGL (Mains) 2020]

Soln:-

I

II

eff. Rate:- 21%

25.44%

given $\rightarrow$ 1050

$$\text{Principal} = \frac{1050}{21} \times 100$$

$$P = 5000$$

$$CI_{II} = 5000 \times \frac{25.44}{100}$$

$$= \boxed{1272} \text{ Ans}$$

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Best of Luck